

Mathematical Foundations of Political Methodology

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Day 8: Confidence intervals

- Interval Estimation
- Hypothesis Testing
- Power and Confidence
- P values

Interval Estimation

- An interval estimator maps a random variable onto an interval, rather than a point.
- If our estimator is $\hat{\theta}$, an interval estimator is written

$$[\hat{\theta}_{lower}, \hat{\theta}_{upper}]$$

- The interval represents the range of possible values within which we are confident to find the true θ .
- This is often called a confidence interval.

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Interval Estimation

Definition

A confidence interval with a confidence level $1 - \alpha$ is an interval estimator $[\hat{\theta}_{lower}, \hat{\theta}_{upper}]$, that is guaranteed to cover the true parameter $100 \times (1 - \alpha)$ of the time.

- Coverage probability describes how likely it is that the interval covers the true parameter value.
- Shorter intervals are better.

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Normal Population with known σ^2

- Suppose we have n draws from a Normal distribution $N(\mu, 2)$

$$\frac{\bar{X}_n - \mu}{2/\sqrt{n}} \sim N(0, 1)$$

- Using the inference from before,

$$F(1.96) = P\left(\frac{|\bar{X}_n - \mu|}{2/\sqrt{n}} < 1.96\right) = .95$$

$$P(-1.96 < \frac{\bar{X}_n - \mu}{2/\sqrt{n}} < 1.96) = .95$$

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- A confidence interval is just a rearrangement:

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$$P(-1.96 * 2/\sqrt{n} < \bar{X}_n - \mu < 1.96 * 2/\sqrt{n}) = .95$$

$$P(\bar{X}_n - 1.96 * 2/\sqrt{n} < \mu < \bar{X}_n + 1.96 * 2/\sqrt{n}) = .95$$

- So the following interval estimator contains the true population μ with probability .95:

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Simulations

```
> library(animation)
> conf.int()
```

Example Normal Population with known σ^2

- Suppose that the WHO is enacting a new Ebola monitoring service at 400 airports which may hold up lines.
- Suppose that we know that the ebola monitoring service has historically had a variance of 9 people per hour and is normally distributed.
- We surveyed 40 of these monitors, and found that the sample had an average speed of 70 people per hour.
- What is the estimate of the speed of the service?
- The following interval **estimator** contains the true population μ with probability .95:

$$[\bar{X}_n - 1.96 * \sigma / \sqrt{n}, \bar{X}_n + 1.96 * \sigma / \sqrt{n}]$$

- The following interval **estimate** has a confidence level of .95:

$$[70 - 1.96 * 3 / \sqrt{40}, 70 + 1.96 * 3 / \sqrt{40}]$$

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The shortest interval

- If $\hat{\theta} \sim N(\mu, \sigma)$

$$[\hat{\theta} - z_{\alpha/2} * \sqrt{\text{var}(\hat{\theta})}, \hat{\theta} + z_{\alpha/2} * \sqrt{\text{var}(\hat{\theta})}]$$

- If $\hat{\theta} \sim t_{n-1}$

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Example Normal Population with unknown σ^2

- Suppose that we are trying to find the average height of a village in Afghanistan.
- Suppose that we know that altitude is normally distributed.
- We surveyed 20 villages, and found that the sample had an average altitude of 3200 meters with a sample variance of 1200 meters.
- What is the estimate of the average altitude in Afghanistan?
- We use the t distribution with 19 degrees of freedom.
- The following interval **estimate** has a confidence level of .95:

$$[3200 - t_{19}(\alpha/2) * \sqrt{1200}/\sqrt{20}, 3200 + t_{19}(\alpha/2) * \sqrt{1200}/\sqrt{20}]$$

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Interpretation

- A 95% confidence interval is an estimate of plausible values for the population parameter.
- It is not a measure of the degree of probability, belief, or support that an unknown parameter value lies in *this* specific interval.
- A particular confidence interval of 95% does not mean that there is a 95% probability of a sample mean from a repeat of the experiment falling within *this* interval.
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Confidence interval for the difference in means with known variances

- Suppose that we are trying to find the average height of a village in Afghanistan relative to a village in Pakistan.
- Suppose that we know that altitude is normally distributed, and the variance is 1200 meters and 200 meters respectively.
- We surveyed 20 villages in Afghanistan and 30 villages in Pakistan, $\bar{X}_A = 3200, \bar{X}_P = 1200$
- What is the estimate of the difference in average altitudes?
- The difference in Normals is normal.

$$\bar{X}_A - \bar{X}_P \sim N \left(\mu_A - \mu_P, \sqrt{\left(\frac{\sigma_A^2}{n_A} + \frac{\sigma_P^2}{n_P} \right)} \right)$$

- The following interval **estimate** has a confidence level of .95:

$$[\bar{X}_A - \bar{X}_P - z_{\alpha/2} * \sqrt{\left(\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y} \right)}, \bar{X}_A - \bar{X}_P + z_{\alpha/2} * \sqrt{\left(\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y} \right)}]$$

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Hypothesis Tests

- Point estimation: what is the most likely value of the parameter?
- Interval Estimation: what is the range of plausible values of the parameter?
- Hypothesis testing: do we have evidence to reject a hypothesized interval of parameter values?
- Failure to find such evidence does not suggest that the hypothesis is true.

Definition of Hypothesis

Definition

A *hypothesis* is a statement about a population parameter

- Does democracy promotion promote peace? (lower rates of conflict among treatment group).
- Do treaties constrain countries? (is behavior distinct among treaty signers?)
- Is support for Trump above 50%?
- Is Ohio State good this year?

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A *Hypothesis Test* compares two mutually exclusive sets of parameters,

- Null hypothesis: H_0
- Alternative hypothesis: H_1

- Does democracy promotion promote peace?
 - H_0 : Democracy promotion does not promote peace (treated and untreated groups are equally conflictual).
 - H_1 : Democracy promotion promotes peace (treated and untreated groups are different).
- Is Ohio State good this year?
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Testing Procedure

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Given a test statistic $t(X)$, a hypothesis testing procedure is a rule that specifies

- *Value of test statistic when we do not reject H_0 : acceptance region*
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- I had a theory that students should study for 10 hours a week.
- Is that theory consistent with practice?
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- Is that theory consistent with practice?
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Error Types

	Fail to Reject H_0	Reject H_0
H_0 True	Correct	Type I Error
H_1 True	Type II Error	Correct

- Type I Error: reject a null that is true

Deciding the homework is too hard when it is not.

- Type II Error: failing to reject a null that is false.

Leaving the homework the same when it is too hard.

The **Size** of test (α) is the probability of Type I Error (false discovery).

The **Power** of test is 1 - the probability of Type II Error (β) (failed discovery).

What is worse: False discovery or missed detection?

What is worse: me having to make homework easier or you having to suffer from continued hard homework.

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Example of Test Statistic and Critical Region

- Set $\mu = 10$
- $t(X) \equiv \frac{\bar{X} - 10}{s/\sqrt{N}} \sim t_{N-1}$
- Critical value (x_0): how long would you have to take on average for me to feel that you are working too hard?
- $P(|t(X)| > x_0) = \alpha$
- $P(t(X) < -x_0) + P(t(X) > x_0) = \alpha$
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Example of Test Statistic and Critical Region

- Set $\mu = 10$ for 22 students.
- Set $\alpha = .05$
- Using the t distribution, x_0 is 2.07
- Suppose the sample mean is 12 hours.
- Suppose the sample standard deviation is 2.6 hours.

$$P(t(X) > x_0)$$

$$\frac{12 - 10}{2.6/\sqrt{22}} > 2.07$$

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$$\text{p-value} = \int_{t(X)}^{\infty} f(x) dx = 0.0008$$

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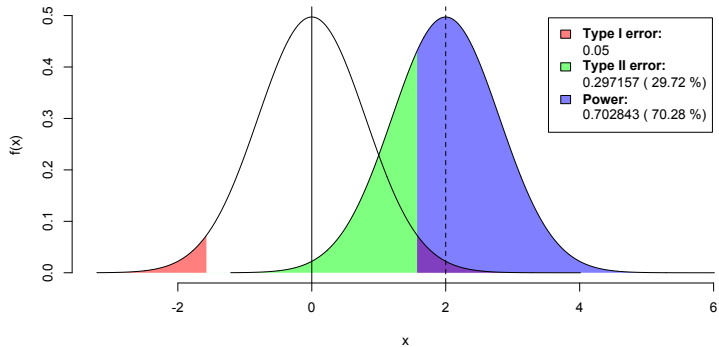
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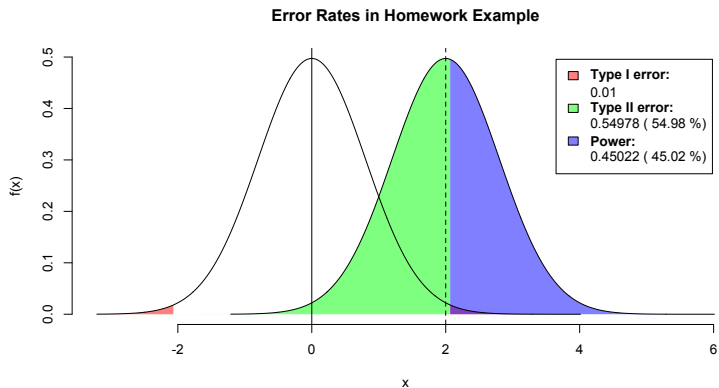
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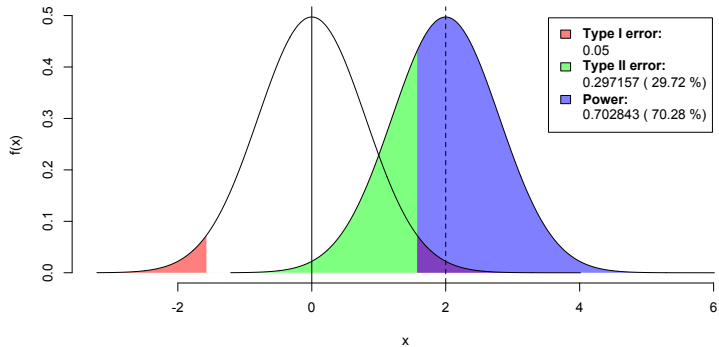
Error Rates in Homework Example



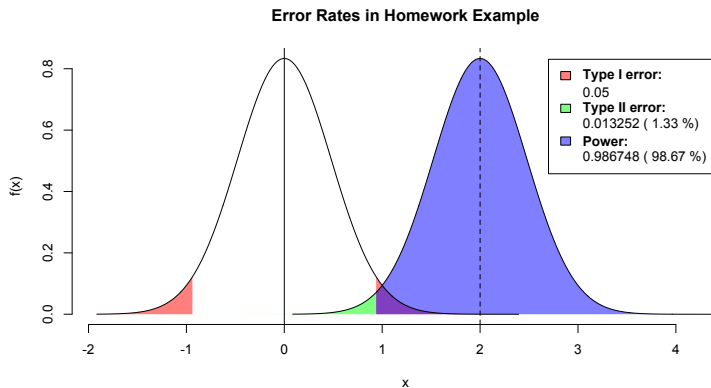
Increase α



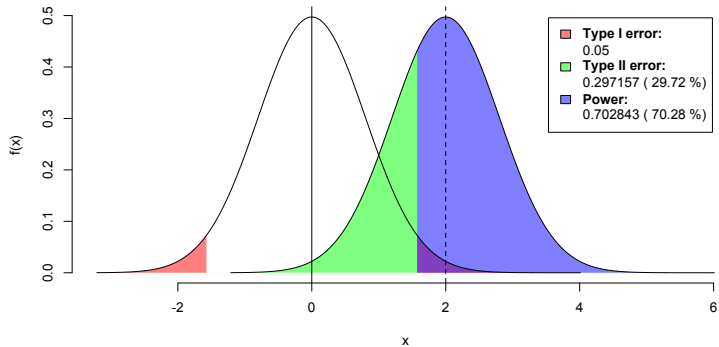
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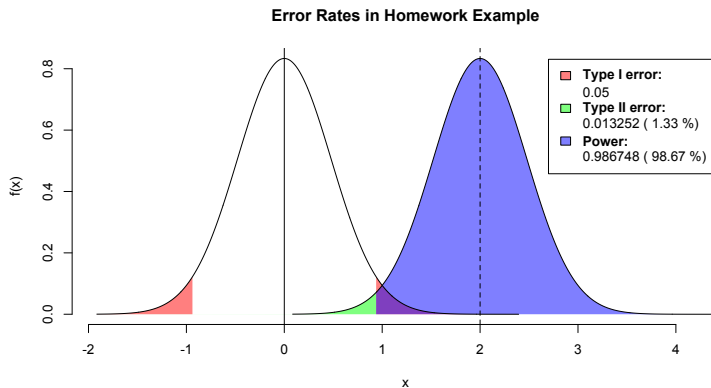
Triple sample size



Error Rates in Homework Example



Decrease σ



Summary

Power will depend on four factors:

- Effect size
- Size of test α
- Sample size
- σ^2

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Confidence Intervals and Hypothesis testing.

- If $\hat{\theta} \sim N(\mu, \sigma)$, the confidence intervals are:

$$[\hat{\theta} - z_{\alpha/2} * \sqrt{\text{var}(\hat{\theta})}, \hat{\theta} + z_{\alpha/2} * \sqrt{\text{var}(\hat{\theta})}]$$

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If $X_1, X_2, X_3, \dots, X_n$ is a random sample from a $N(\mu, \sigma^2)$ data, $\frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ has a student's t distribution with $n - 1$ degrees of freedom.

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What to do?

- Report more information about the distribution.
- Design studies so that you care about the estimates, not just rejecting a null hypothesis.